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The many facets of facial interactions in mammals

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Facial interactions are prominent behaviors in primates. Primate facial signaling, which includes the expression of emotions, mimicking of facial movements, and gaze interactions, is visually dominated. Correspondingly, in primate brains an elaborate network of face processing areas exists within visual cortex. But other mammals also communicate through facial interactions using additional sensory modalities. In rodents, multisensory facial interactions are involved in aggressive behaviors and social transmission of food preferences. The eusocial naked mole-rat, whose face is dominated by prominent incisors, uses facial aggression to enforce reproductive suppression. In burrow-living mammals like the naked mole-rat in particular, and in rodents in general, somatosensory face representations in cortex are enlarged. Diversity of sensory domains mediating facial communication might belie underlying common mechanisms. As a case in point, neurogenetics has revealed strongly heritable traits in face processing and identified gene defects that disrupt facial interactions both in humans and rodents.

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Introduction

Since the pioneering nineteenth century work of Duchenne de Boulogne [1], the mechanisms of facial signaling in primates have been a focus of experimental neuroscience, which, in the past decades, has provided detailed information about the neural processes underlying primate face perception. While facial interactions in primates are easily recognized by researchers belonging to a primate species, only in recent years has it become clear that other mammals engage in elaborate facial interactions as well. Facial interactions are not stereotyped across mammals but are mediated through behaviors and sensory routes commensurate with the general motor and sensory specializations of each species. Here we review commonalities and differences in the anatomy of faces, facial behaviors, neural mechanism and neurogenetics of mammalian facial interactions. We suggest that a neuroethological framework will help unifying the diverse research approaches and make results from one species productive for research on another, thus creating the new field of facial neuroscience.

Primate and rodent faces

Mammals (Figure 1), in contrast to other terrestrial vertebrates, possess facial muscles that directly insert at the outer ears, the vibrissae, and the skin, allowing for active signaling and thus direct facial communication.

Primate faces

Primate species are all highly social. Their social behaviors frequently involve facial communication that can include tactile signals (as in kissing) or scents, but is primarily mediated through auditory and visual channels. Supporting these forms of communication, the facial musculature of primates is complex, consisting of numerous muscles with discrete attachment sites. In the rhesus monkey (Figure 1a), 24 facial muscles have been identified [2•], a number close to the 23 found in chimpanzee and human (Figure 1b) [3]. In all three species, it has been possible to directly map activity of individual facial muscles, elicited by intramuscular electrical stimulation, to facial movements [4], and compositions of these individual muscle movements explain naturally occurring facial expressions [5]. These species similarities support Darwin's hypothesis that human facial expressions are an evolutionary heritage [6].

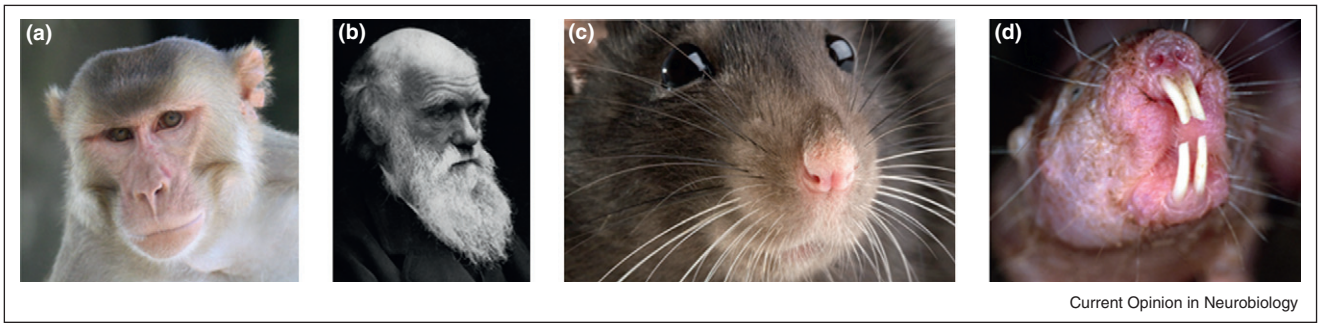
Rodent faces

Rodent facial signaling includes tactile interactions by highly mobile whiskers (Figure 1c) supported by an elaborate musculature [7], visual signals such as ear wiggling, pheromonal signals from cheek glands [8], and facially transmitted combined olfactory-gustatory signals [9]. This genuinely multisensory facial signaling, which distinguishes rodents from visually dominated primates, requires close facial contact between animals. Close contacts arise, because many rodents are small, nocturnal and live in burrows. Facial contacts are prominent in experimental burrow habitats in rats [10] and are of utmost importance in entirely burrow-living (fossorial) mammals like the naked mole-rat (Figure 1d).

Primate facial interactions and their neural mechanisms

Primates rapidly recognize facial signals, which might combine with a specific body posture or vocalization,

Figure 1



Rodent and primate faces. **(a)** A face of macaque monkey. Gaze is a prominent signal in primates. **(b)** The human face differs in many significant ways from other primate faces. Derived features include white sclerae, hair-free skin and red lips. **(c)** A rat face. Many parts of the rat face (the highly mobile whiskers, cheek glands, saliva from the mouth) have been shown to be involved in social signaling. **(d)** The naked mole-rat's face is dominated by its huge incisors. In this eusocial mammal pheromonal signaling appears to be reduced, but antagonistic tactile facial signals such as incisor fencing and shoving are prominent. Modified from [81].

and often reciprocate with their own facial response, thus engaging in a facial interaction. Prominent examples are contagious yawning [11], exchange of facial expressions, gaze following, and facial mimicking.

Facial expression

Facial expressions can signal the emotional state of an animal. Despite similarity of facial expressions across species [12], the same facial expression might signal different emotional states in closely related species [13]. This raises the question of whether the patterns of muscle contractions forming a facial expression are arbitrary or serve an adaptive function [6]. One kind of adaptive advantage a facial expression, for example, fear, can carry, is an alteration of sensory acquisition, for example, through widening of the eyes, which might be advantageous for acquiring more information about the source of fear. Indeed, the muscle contraction patterns of human fearful and disgusted expressions enhance and reduce acquisition of sensory information, respectively [14•]. While little is known about the neurophysiology of facial expression generation, facial expression recognition has been intensively studied. The discrimination of human basic facial expressions relies on those parts of the face that are the physically most distinctive for a given expression, for example, the smiling mouth in 'happy' or the wide-opened eyes in 'fear' [15•]. Use of these diagnostic facial features in fear recognition is impaired after lesion of the amygdala [16], a brain region known from single unit recordings to contain neurons selective for facial expressions [17]. Facial expression-selective neurons have further been found in inferotemporal cortex [18], orbitofrontal cortex [19], and the pulvinar [20].

Gaze following

Gaze, head, and body direction can reveal an animal's interests and intentions. Gaze direction is particularly important in primates for anatomical reasons [21]: First, head orientation is discriminated easily only in heads with large protrusions like snouts, features that are reduced in many primate species (Figure 1b). Second, in primates gaze direction cannot be inferred from head orientation, since primate facial musculature allows for fine control of eye movements independently of head orientation. Third, gaze direction can be easily inferred from particularly visible eyes like those of humans [22] — which might in fact have been the reason for the white sclera of the human eye to evolve [23]. Primates recognize gaze in others to orient their own attention [24]. Gaze following has a fast, reflexive component [25] and occurs even when detrimental to task performance [26]. The neural networks mediating this behavior have to link gaze direction analyzers to attentional control regions. Gaze-selective neurons and areas have been found inside the superior temporal sulcus (STS) and the amygdala [27–29], structures whose lesioning impairs gaze following [30,31]. A recent electrophysiological study [32•] found a putative neural correlate of attentive orienting during gaze following: neurons in the lateral intraparietal area, thought to represent the focus of attention and movement intentions, were modulated by perceived gaze.

Facial mimicking

Newborn primates are, amazingly, able to imitate facial gestures and expressions. Facial mimicking has been found in several primate species including humans [33], chimpanzees [34], and macaques [35••] for a limited period of time after birth. The neural underpinnings of facial mimicking are unknown, but facial mirror-neurons [36] might be involved.

Neural specializations for face recognition

Primate facial interactions are based on visual face-recognition. The primate brain exhibits at least four levels of specialization for face recognition. First, face-selective neurons were found in the macaque temporal lobe

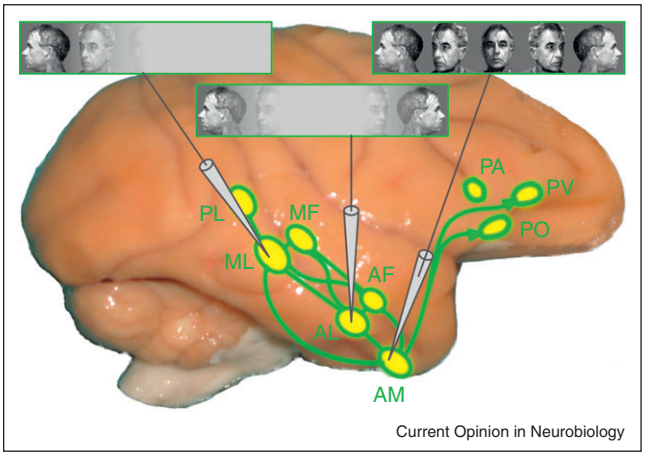
[37,38]. Second, face-selective columns were described [39]. Third, face-selective cortical patches of a few millimeters diameter were discovered in macaques [40–43], similar to the larger face-selective areas in humans [44]. Face patches were found to be almost exclusively composed of face-selective neurons [45,46]. Fourth, face patches and face areas are selectively coupled to form face-processing networks [47,48] (Figure 2). Spatial organization of face-cells correlates with functional specialization: while cells within a face patch seem to encode faces in similar ways, for example, via tuning to facial parts [49], face-cells in different brain regions are specialized for different aspects of faces [18,46,50]. Similarly in humans, face areas are differentially specialized [51–53]. These different levels of organization have been linked to face-recognition abilities through comparison of neural tuning to psychophysics [54], correlation of face-selectivity and face-recognition abilities [55], co-variation analysis of activity in face areas and face-recognition performance [56], causation of face detection biases through electrical stimulation in clusters of face cells [57], and transcranial magnetic stimulation induced face-recognition deficits [58].

Rodent facial interactions

Rat active sensing, facial touch, and the role of whiskers in aggression

Rats possess an elaborate whisker active sensing system that participates in social interactions. Specifically, facial touch is an amazing behavior involving intense mutual facial whisking [59•]. When rats are introduced to each other, they engage in anogenital sniffing and facial interactions. Anogenital sniffing predominates in the first minutes. Thereafter, facial interactions occur more frequently. As rats approach each other, whisking amplitude decreases and whisker set angle increases. Contact is made first with the noses, while the rats align face-to-face. Whisker trimming disrupts this alignment and diminishes facial interactions. Whisker trimming also increases aggressive behaviors such as biting and strongly suppresses boxing behavior [60,61]. Additional observations underscore the importance of whiskers in social signaling. Mice perform mutual whisker removal, the so-called 'barbering behavior', tightly related to dominance [62]. In aggressive interactions, rats keep whiskers protracted and move them with higher amplitude than during non-aggressive interactions. These findings suggest that rodents use their whiskers to display facial expressions [59•].

Figure 2



Macaque cortical face-processing network. Lateral view of macaque brain with approximate locations of six temporal and three prefrontal face patches (green/yellow ellipses, [40–42]) and known connections ([47] and Moeller S, Freiwald WA, Tsao DY: A direct link between face-selective regions in temporal and prefrontal cortex. *Soc. Neurosci. Abstr.* 2008, 261.5). Functional specialization of face patches ([46]) is indicated by schematic recording electrodes and sample tuning to head orientation in middle face patches (top left panel), face patch AL (middle) and face patch AM (top right). Cells in the middle face patches are tuned to one head orientation, cells in AL to multiple, typically mirror-symmetric head orientations, and cells in AM are almost invariant to head orientation.

Facial interactions and the social transmission of food preferences

In rodents, a demonstrator's food consumption can be sensed by observers and facilitates their later consumption of novel diets or the searching out of distant food sources [63,64]. Early work emphasized the purely olfactory transmission of preferences without demonstrator–observer interactions [63,64]. Subsequent studies, however, identified a strong influence of facial interactions [65] with live demonstrators in social transmission. Observers detect the combination of novel food smells and carbon sulfides contained in rodent saliva [9]. Recent work identified the olfactory receptors for carbon disulfides and demonstrated their involvement in transmission of food preferences [66•].

Social suppression by facial aggression in the naked mole-rat

The eusocial naked mole-rat shows a unique set of aggressive facial behaviors. Antagonistic facial behaviors include open-mouth gaping, incisor fencing, biting, tugging, and shoving [67]. Shoving is a particularly frequent behavior directed from the queen to high-ranking colony members [68], lazy workers [69], and pups. Queen removal experiments suggest that shoving enforces reproductive suppression of workers [68].

Primary sensory cortices, the hippocampus, and facial interactions

The neural analysis of rodent facial interactions is in its infancy and is thus difficult to compare to that of primates. However, comparison of the earliest stages of sensory cortical processing and of high-level representations in the hippocampus can serve as starting points for such a comparison.

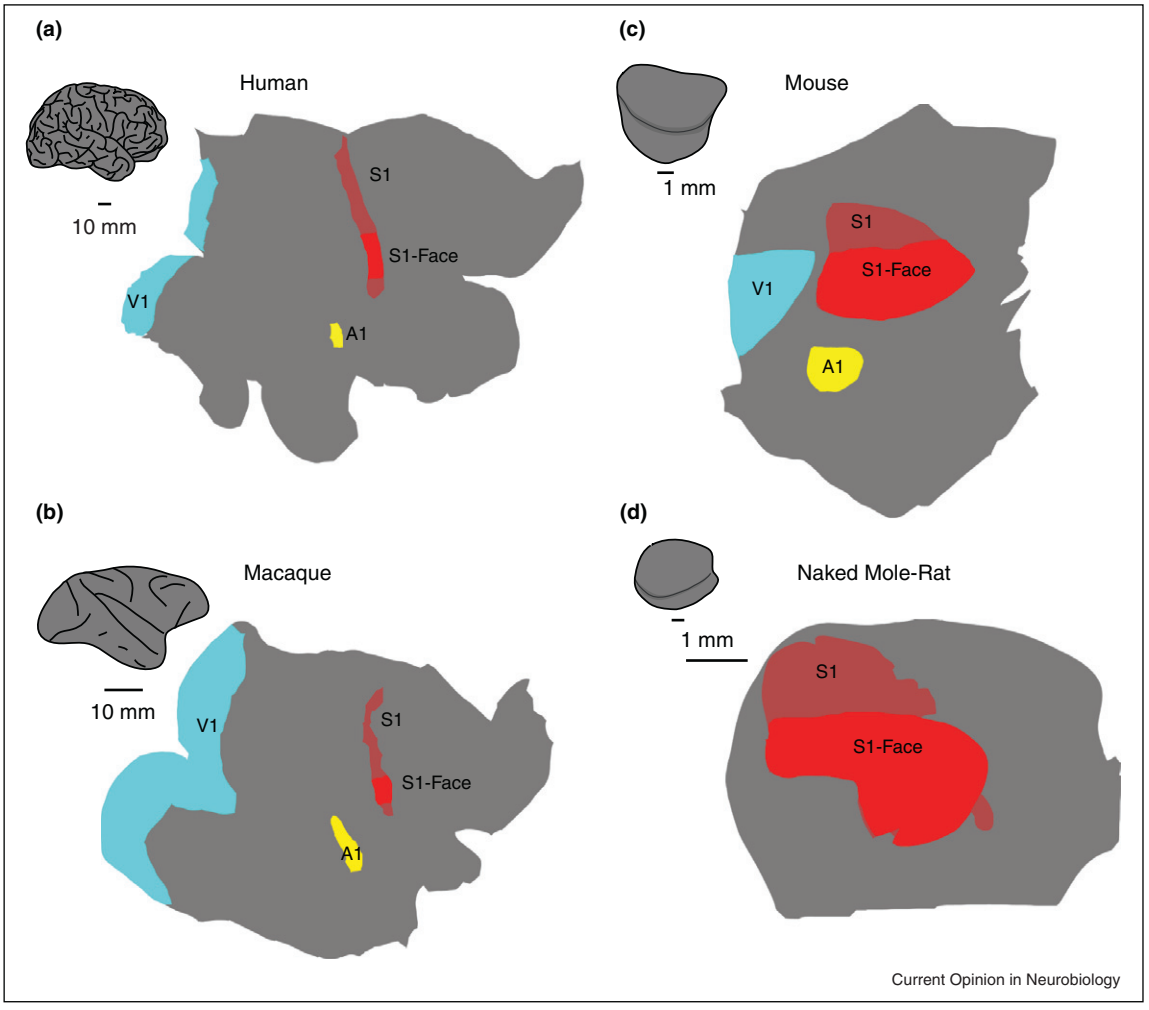
Primary sensory cortices and the modality of facial signaling

Primary sensory cortices in rodent and primate species are of strikingly different sizes (Figure 3). Large visual cortices are thought to go along with diurnal arboreal life styles [70], typical for many primate species, while large somatosensory cortices are found in nocturnal [71] and fossorial animals [72], such as many rodents. Thus ecological factors might also explain the predominance of visual facial signaling in primates and tactile facial behaviors in rodents, as well as the large differences in primary somatosensory representations of faces, which occupy ~0.5% of cortical surface area in macaques, ~0.7% in humans, ~10% in mice, and ~20% in naked mole-rats. The sheer size of the rodent somatosensory face representation necessitates an investigation into its contribution to the complex facial interactions described above. Even though the somatosensory face representation in primates is relatively smaller than that in rodents, it is large in absolute size, and cortical face representations of primates and rodents show remarkable similarities in their fine structure and modular architecture [73].

Expansion of the face representation and social suppression in naked mole-rats

In the mole-rat face representation, the teeth (which figure prominently in digging) are emphasized [74]. A similar expansion of the somatosensory face and tooth cortex is seen in unrelated and solitary subterranean mole-rats [72]. Thus face expansion in mole-rats is probably related to subterranean life-style and not to social interactions. We hypothesize that neural face expansion preceded the evolution of eusociality in mole-rats. This neural face expansion, then, likely is the key reason why

Figure 3



Flat maps of rodent and primate cortices. Side views and flat maps of right cortical hemispheres of **(a)** human, **(b)** macaque monkey, **(c)** mouse and **(d)** naked mole-rat with superimposed outlines of primary sensory cortices (S1 = primary somatosensory area, V1 = primary visual cortex, A1 = primary auditory cortex. In all four species the flat map has been normalized to the same area. In monkey and human the outline of somatosensory cortex refers to putative areas 3a and 3b. In (a) and (b) scale bar apply both to the side view and the flat map, whereas in (c) and (d) different scales apply with the lower one referring to the flat map. Data have been modified from [97–100].

facial interactions are so prominent in this species. Naked mole-rats are able to dominate conspecifics through facial behaviors (shoving). Consistent with facially mediated dominance, reproductive suppression in naked mole-rats does not appear to be mediated by primer pheromones [75]. The vomeronasal organ, key mediator of pheromonal transmission in rodents [76], is reduced in naked mole-rats [77•]. As pheromones mediate reproductive suppression in other mammals [78,79] and insects, the mole-rat is highly unusual in this regard.

In naked mole-rat colonies, facial aggression shapes social structure, and social status (queen vs. worker) rather than gender appears to shape the mole-rat's brain. Holmes *et al.* [80••] analyzed a set of subcortical nuclei involved in reproductive behavior. While in most mammals the size of these nuclei is sexually dimorphic, in naked mole-rats, it instead depends on social status and not gender. The loss of neural sexual dimorphisms parallels the loss of sexually dimorphic genitalia in the evolution of mole-rat eusociality.

Hippocampus and social concepts

In the human hippocampus, neurons selective for familiar individuals have been found [81], thought to represent conceptual knowledge relevant for social interactions. Subsequently, hippocampal activity in interacting rats has been investigated with electrophysiology and immediate-early-gene expression mapping [82•,83•]. Rodent hippocampal neurons hardly responded to and did not discriminate between conspecifics [82•]. Even during facial interactions place cell activity was only modestly modulated by conspecifics. Similarly, most [84,85], but not all [86] rodent lesion studies argue against a social role of the rodent hippocampus.

Neurogenetics of facial interactions

Genetics of face processing

Are neural specializations for faces present from birth, or do specializations develop inside a general purpose object recognition system that frequently encounters faces? Innate mechanisms are suggested by facial mimicking in newborns [35••] and preferential looking at faces over non-face objects in macaque monkeys raised without exposure to faces [87••]. Yet the same monkeys experienced declined recognition abilities for faces of a primate species they did not encounter.

Two twin studies [88••,89••] provided stunning evidence for a strong genetic component in human face processing. Face processing abilities were much more strongly correlated in monozygotic twins than in same-sex dizygotic twins. A genetic basis specifically for face recognition is the likely cause, since face recognition performance correlated only weakly with other visual and cognitive abilities. Recently identified heritable forms of face-blindness (prosopagnosia) [90,91] also point to a specific genetic substrate of face processing. Facial processing deficits were remarkably specific in some cohorts, impairing face identification, without compromising facial attractiveness judgments [92••].

Disturbed facial interactions in autism

Autism Spectrum Disorders (ASDs) are heritable conditions associated with deficits in social communication, impaired language skills and stereotypy. Converging evidence from monogenetic defects and genetic screens identified two major pathways contributing many autistic phenotypes [93]. One pathway leads to macrocephaly and ASD through abnormal growth rates. Other pathways affect synaptic function through alterations in synaptic proteins such as neuroligins [94•], shank-proteins [95••] and neurexins and leads to mental retardation, autism or Asperger syndrome. Many autistic patients suffer from heritable face recognition defects. Mice with gene defects involved in autism also show aberrant social behavior [94•] and provided intriguing similarities in facial interactions of mice and men. The Phelan-McDermid syndrome is a monogenetic form of autism, in which the synaptic scaffolding protein Shank 3 is affected. Patients often show poor eye contact [96] similar to other autistic patients. Remarkably, the corresponding mouse model also engages in fewer facial contacts [95••]. To understand how alterations of synaptic proteins might cause specific impairments of facial behaviors, will require a synthesis of molecular neuroscience with primate and rodent facial neurobiology.

Neurogenetics might also reveal whether facial interactions in mammals evolved from the same neural machinery or not. A common mammalian predisposition for facial interaction stems from suckling, by which all mammals establish a first, highly significant and intense social interaction.

Conclusion

The discovery of face cells spawned one of the most fascinating chapters of neuroscience. With increased recognition that facial interactions are not restricted to primates, studies on visual face perception should be complemented by broad neuroethological work on facially interacting mammals. Rodent facial interactions are elaborate behaviors, which are particularly highly developed in the eusocial naked mole-rat. Powerful neurogenetic tools spur interest in combining rodent and primate face neuroscience.

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